Phoebe Mihevc

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EDUCATION

University of Illinois at Urbana-Champaign, Grainger College of Engineering **Expected May 2027**
Bachelor of Science in Mechanical Engineering GPA: 3.52/4.00
Relevant Coursework: Design for Manufacturability, Additive Manufacturing & Product Design, Bioinspired Design

WORK EXPERIENCE

Corporate Procurement Intern, Caterpillar, Inc. – **Peoria, IL** **May 2026 – August 2026**
• Leveraged generative AI models alongside Power Query and Python in Excel to create an inactive-part reporting tool to reduce manual effort and mitigate risk; identified that ~40% of active inventory could be reclassified
• Designed and deployed an inventory management system in Excel for a new product introduction (NPI) build facility, establishing clear visibility into component availability, usage, and build readiness to proactively identify and address material shortages, significantly reducing build delays
• Served as a key liaison between engineering teams and the build technician for NPI efforts, ensuring proper implementation of design changes, communicating points of contention, providing regular status updates, and coordinating procurement of missing components

Maintenance Engineering Intern, Hormel Foods – **Wichita, KS** **May 2025 – August 2025**
• Managed maintenance operations and a team of mechanics across a major plant division, including planning and scheduling preventative maintenance, performing Root Cause Analysis to address mechanical deficiencies, and coordinating cross-departmentally to minimize machine downtime and increase production efficiency
• Managed multiple projects and coordinated with external contractors to obtain quotes, communicate plant requirements, and ensure informed decision making
• Mapped a facility-wide piping network and executed iterative CAD redesigns in AutoCAD to minimize cost, improve fluid delivery efficiency, prevent future corrosion, and enable accurate budgeting
• Directed the design of a heat exchanger system, collaborating with contractors to understand mechanics, reduce mechanical breakdown, and ensure compliance with OSHA standards and changing plant conditions

RESEARCH

Undergraduate Researcher, Glumac Group, UIUC – **Champaign, IL** **September 2024 – Present**
• Heavily supported the development of a novel, high-sensitivity spectroscopic analysis system by establishing DAQ hardware and software connections for a 200-photomultiplier tube (PMT) module array to capture real-time data
• Conducted experiments using a heterogeneous combustion shock tube for extreme condition diagnostics
• Modeled custom hardware components and engineering drawings in Fusion 360

PROJECTS

5-Speed Robotic Car, *Mechanical Design Course* **January 2026 – May 2026**
• Collaborated in a 4-person team to engineer and manufacture a remote-controlled robotic car, meeting strict speed, load, and performance benchmarks
• Designed a remote-controlled 5-speed transmission using Fusion360 and Arduino, achieving a gearbox efficiency of 93% by minimizing friction and applying DFM principles to optimize component meshing
• Diagnosed and resolved system-wide integration issues through iterative physical testing and tolerance analysis

1-DOF 6-Legged Dispenser-Walker Robot, *Mechanical Design Course* **August 2025 – December 2025**
• Designed and fabricated a robot that walks 6m/min and dispenses a small payload every 2m
• Engineered component mounts and housings, and an anti-jamming dispensing mechanism, using physical testing and rapid prototyping to eliminate feed jams and ensure consistent payload delivery
• Performed root cause analysis to resolve gait instability and misalignment

Six-Axis Robot Arm, *American Society of Mechanical Engineers (ASME)* **January 2025 – May 2025**
• Used Fusion 360 to design a casing for a 6-axis robotic arm, regularly refining the design to accommodate ongoing modifications while ensuring ease of assembly and aesthetic quality

SKILLS

Software: Fusion360, Microsoft Office Suite, Excel, Google Workspace/Suite, Cura, CoMPASS, Power Query, nTop
Tools: 3D Printing, Power Tools, Woodworking Tools (Drill Press, Table Saw, etc.), Manual Mill and Lathe